

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Soil Input	Accepts soil & climate values	React UI	Done	1
2	Input Validation	Validates user inputs	Flask Backend	Done	1
3	Crop Prediction	Predicts suitable crop	Random Forest Model	Done	1
4	REST API	Connects frontend & backend	Flask API	Done	1
5	Result Display	Shows predicted crop	React UI	Done	1
6	Dataset Loading	Shows predicted crop	Backend	Done	1
7	Model Loading	Loads trained .pkl model	Backend	Done	1
8	Responsive UI	Responsive web interface	React	Done	1

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Prediction Form, Result Card
2	Backend / Logic	3	Validation, Prediction, Model Loading
3	Database / Storage	1	Crop Dataset, ML Model
4	API / Integration	1	Flask REST API
5	Security / Authentication	1	Input Validation